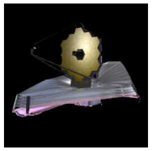

# 2562 - Dust Settling and Grain Evolution in Edge-on Protoplanetary Disks

Cycle: 1, Proposal Category: GO

**INVESTIGATORS**


| *Name* | *Institution* |
|---|---|
| **Dr. Francois Menard (PI) (ESA Member)** | **Institut de Planetologie et d'Astrophysique de Grenoble** |
| Dr. Karl Stapelfeldt (CoI) (CoPI) (US Admin CoI) (Contact) | Jet Propulsion Laboratory |
| Dr. Gaspard Duchene (CoI) | University of California - Berkeley |
| Dr. Marion Villenave (CoI) | Jet Propulsion Laboratory |
| Dr. Schuyler G. Wolff (CoI) | University of Arizona |
| Dr. Marshall Perrin (CoI) | Space Telescope Science Institute |
| Dr. Christophe Pinte (CoI) | Monash University |
| Dr. Deborah Padgett (CoI) | Jet Propulsion Laboratory |


**OBSERVATIONS**

| *Folder* | *Observation* | *Label* | *Observing Template* | *Science Target* |
|---|---|---|---|---|
| Tau0420201 | | | | |
| | 1 | Tau042021 - NIRCam | NIRCam Imaging | (1) 2MASS-J04202144+2813491 |
| | 2 | Tau042021 - MIRI | MIRI Imaging | (1) 2MASS-J04202144+2813491 |
| IRAS 04302+2247 | | | | |
| | 3 | IRAS04302 - NIRCam | NIRCam Imaging | (2) IRAS-04302+2247 |
| | 4 | IRAS04302 - MIRI | MIRI Imaging | (2) IRAS-04302+2247 |
| HH30 | | | | |
| | 5 | HH30 - NIRCam | NIRCam Imaging | (3) HH-30-NIRCAM |
| | 6 | HH30 - MIRI | MIRI Imaging | (4) HH-30-MIRI |
| HH30 - repeat | | | | |
| | 15 | HH30 - NIRCam - repeat | NIRCam Imaging | (3) HH-30-NIRCAM |

| *Folder* | *Observation* | *Label* | *Observing Template* | *Science Target* |
|---|---|---|---|---|
| | 16 | HH30 - MIRI - repeat | MIRI Imaging | (4) HH-30-MIRI |
| Oph163131 | | | | |
| | 7 | Oph163131 - NIRCam | NIRCam Imaging | (5) 2MASS-J16313124-2426281 |
| | 8 | Oph163131 - MIRI | MIRI Imaging | (5) 2MASS-J16313124-2426281 |

**ABSTRACT**

Young, edge-on circumstellar disks are uniquely valuable laboratories for the study of planet formation. With the central star occulted from direct view, the disk is clearly seen as a central dust lane flanked by reflected light from its upper and lower surfaces. The detailed morphology and chromaticity of these nebulae provides crucial information on disk vertical structure and the properties of its constituent dust grains. Spectral energy distributions and very limited groundbased imaging have shown that edge-on protoplanetary disks continue to be dominated by scattered light even out to wavelengths of 20 microns. JWST imaging of these targets therefore offers the unique opportunity to probe the disk interior between the optical scattered light surface seen with HST and the cold midplane emission recently revealed by ALMA. We propose broad-band NIRCam and MIRI imaging of four edge-on protoplanetary disks spanning a range of evolutionary states. The targets are among the largest in angular size of their class and thus should be vertically resolved by JWST even at 20 microns. The images will reveal the wavelength evolution of both the dust lane thickness and the strength of forward scattering, which when interpreted by model fitting will allow us to derive the grainsize as a function of height in the disk and thus the extent of dust vertical settling that has taken place. This project will empirically quantify for the first time how the dust concentration increases toward the disk midplane (a necessary condition to efficiently form planetesimals), leaving a legacy of fundamental importance for our understanding of protoplanetary disk evolution.

**OBSERVING DESCRIPTION**

We will observe each of our four targets in five filters to probe the evolution of the disk scattered light nebulosity with wavelength. As the target stars are variable, we require all the observation of each target to execute consecutively without interruption.

The NIRCam observations consist of a F200W/F444W image pair. All of our primary targets are smaller than 20 arcsec in extent. We will image with the full array without large dithers, but with subpixel dithers at 4 positions to reject bad pixels and build up subsampled images. These images are designed to sample the wavelength evolution of the nebula structure, taking advantage of JWST's stability to make the best strehl ratio images ever obtained for these targets in K and M band. The S/N and PSF quality are crucial for good modeling results: the brightness gradients of the two disk nebulae as they fade into the central dark lane are the primary observables constraining the disk scale height. The total charged time for the NIRCam observations is 3.7 hours.

The MIRI observations consist of F770W, F1280W. and F2100W images. The BRIGHTSKY subarray will be used to limit saturation on the thermal background in the F2100W filter; its 56" field of view is sufficient to cover our targets. The observations will be four-point dithered with the extended source dither scale. Due to the variation in integrated brightness and nebular area (by as much factors of 5 between targets), the exposure times must be tailored for each. As the faintest and largest disk, Tau042021 requires almost three times as much MIRI integration time as the smallest target Oph163131. Despite the higher backgrounds, the F2100W exposures are shorter than those at F1280W due to the source emission spanning fewer diffraction-limited resolution elements. The total charged time for MIRI observations is 7.1 hours.

**Fixed Targets**

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | 2MASS-J04202144+2813491 | RA: 04 20 21.4428 (65.0893450d)<br>Dec: +28 13 49.17 (28.23032d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |
| *Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br><br>K= 14.05<br>Category=Star<br>Description=[Circumstellar disks, Herbig-Haro objects, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]<br>Extended=YES* | | | | |
| (2) | IRAS-04302+2247 | RA: 04 33 16.5010 (68.3187542d)<br>Dec: +22 53 20.40 (22.88900d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |
| *Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br><br>K= 11.7<br>Category=Star<br>Description=[Circumstellar disks, Herbig-Haro objects, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]* | | | | |
| (3) | HH-30-NIRCAM | RA: 04 31 39.6595 (67.9152479d)<br>Dec: +18 13 5.42 (18.21817d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |
| *Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=Star<br>Description=[Circumstellar disks, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]* | | | | |
| (4) | HH-30-MIRI | RA: 04 31 37.4719 (67.9061329d)<br>Dec: +18 12 24.48 (18.20680d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |
| *Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=Star<br>Description=[Circumstellar disks, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]* | | | | |
| (5) | 2MASS-J16313124-2426281 | RA: 16 31 31.2518 (247.8802158d)<br>Dec: -24 26 28.47 (-24.44124d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |
| *Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br><br>K= 12.63<br>Category=Star<br>Description=[Circumstellar disks, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]* | | | | |

## Observation

**Proposal 2562, Observation 1: Tau042021 - NIRCam**

Tue Aug 29 22:01:08 GMT 2023

**Diagnostic Status: Warning**

Observing Template: NIRCam Imaging

## Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | 2MASS-J04202144+2813491 | RA: 04 20 21.4428 (65.0893450d)<br>Dec: +28 13 49.17 (28.23032d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |

*Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.*

*K= 14.05*
*Category=Star*
*Description=[Circumstellar disks, Herbig-Haro objects, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]*
*Extended=YES*

## Template

| Module | Subarray | Target Placement |
|---|---|---|
| B | FULL | Module Gap |

## Dithers

| # | Primary Dither Type | Primary Dithers | Subpixel Dither Type | Dither Size | Subpixel Positions |
|---|---|---|---|---|---|
| 1 | NONE | | STANDARD | | 4 |

## Spectral Elements

| # | Short Filter | Long Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|
| 1 | F200W | F444W | BRIGHT2 | 5 | 3 | 12 | 4 | 1374.307 | |

## Special Requirements

Offset 44.53572562082701 arcsec, -15.858058198923379 arcsec

Sequence Observations 1, 2, Non-interruptible

## Observation

**Proposal 2562, Observation 2: Tau042021 - MIRI**

Tue Aug 29 22:01:08 GMT 2023

**Diagnostic Status: Error**

Observing Template: MIRI Imaging

## Diagnostics

(Tau042021 - MIRI (Obs 2)) Error (Form): Permission has not been granted for this program to use Special Requirement 'No Parallel Attachments'.

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | 2MASS-J04202144+2813491 | RA: 04 20 21.4428 (65.0893450d)<br>Dec: +28 13 49.17 (28.23032d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |

*Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.*

*K= 14.05*
*Category=Star*
*Description=[Circumstellar disks, Herbig-Haro objects, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]*
*Extended=YES*

## Template

| Subarray |
|---|
| BRIGHTSKY |

## Dithers

| # | Dither Type | Starting Point | Number of Points | Points | Starting Set | Number of Sets | Optimized For | Direction | Pattern Size |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 4-Point-Sets | | | | 4 | 1 | EXTENDED SOURCE | POSITIVE | DEFAULT |

## Spectral Elements

| # | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | F770W | FASTR1 | 60 | 5 | 1 | Dither 1 | 4 | 20 | 1052.18 | |
| 2 | F1280W | FASTR1 | 30 | 12 | 1 | Dither 1 | 4 | 48 | 1284.076 | |
| 3 | F2100W | FASTR1 | 15 | 24 | 1 | Dither 1 | 4 | 96 | 1325.609 | |

## Special Requirements

No Parallel Attachments

Sequence Observations 1, 2, Non-interruptible

## Observation

**Proposal 2562, Observation 3: IRAS04302 - NIRCam**

Tue Aug 29 22:01:08 GMT 2023

**Diagnostic Status: Warning**

Observing Template: NIRCam Imaging

## Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (2) | IRAS-04302+2247 | RA: 04 33 16.5010 (68.3187542d)<br>Dec: +22 53 20.40 (22.88900d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |

*Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.*

*K= 11.7*
*Category=Star*
*Description=[Circumstellar disks, Herbig-Haro objects, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]*

## Template

| Module | Subarray | Target Placement |
|---|---|---|
| B | FULL | Module Gap |

## Dithers

| # | Primary Dither Type | Primary Dithers | Subpixel Dither Type | Dither Size | Subpixel Positions |
|---|---|---|---|---|---|
| 1 | NONE | | STANDARD | | 4 |

## Spectral Elements

| # | Short Filter | Long Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|
| 1 | F200W | F444W | BRIGHT2 | 5 | 2 | 8 | 4 | 901.889 | |

## Special Requirements

Offset 34.60783563455311 arcsec, -25.368706229956377 arcsec

Sequence Observations 3, 4, Non-interruptible

## Observation

**Proposal 2562, Observation 4: IRAS04302 - MIRI**

Tue Aug 29 22:01:08 GMT 2023

**Diagnostic Status: Error**

Observing Template: MIRI Imaging

## Diagnostics

(IRAS04302 - MIRI (Obs 4)) Error (Form): Permission has not been granted for this program to use Special Requirement 'No Parallel Attachments'.

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (2) | IRAS-04302+2247 | RA: 04 33 16.5010 (68.3187542d)<br>Dec: +22 53 20.40 (22.88900d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |

*Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.*

*K= 11.7*
*Category=Star*
*Description=[Circumstellar disks, Herbig-Haro objects, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]*

## Template

| Subarray |
|---|
| BRIGHTSKY |

## Dithers

| # | Dither Type | Starting Point | Number of Points | Points | Starting Set | Number of Sets | Optimized For | Direction | Pattern Size |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 4-Point-Sets | | | | 4 | 1 | EXTENDED SOURCE | POSITIVE | DEFAULT |

## Spectral Elements

| # | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | F770W | FASTR1 | 45 | 4 | 1 | Dither 1 | 4 | 16 | 633.385 | |
| 2 | F1280W | FASTR1 | 120 | 4 | 1 | Dither 1 | 4 | 16 | 1671.721 | |
| 3 | F2100W | FASTR1 | 5 | 16 | 1 | Dither 1 | 4 | 64 | 328.806 | |

## Special Requirements

No Parallel Attachments

Sequence Observations 3, 4, Non-interruptible

## Observation

**Proposal 2562, Observation 5: HH30 - NIRCam**

Tue Aug 29 22:01:08 GMT 2023

**Diagnostic Status: Warning**

Observing Template: NIRCam Imaging

## Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (3) | HH-30-NIRCAM | RA: 04 31 39.6595 (67.9152479d)<br>Dec: +18 13 5.42 (18.21817d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |

*Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.*
*Category=Star*
*Description=[Circumstellar disks, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]*

## Template

| Module | Subarray | Target Placement |
|---|---|---|
| B | FULL | Module Gap |

## Dithers

| # | Primary Dither Type | Primary Dithers | Subpixel Dither Type | Dither Size | Subpixel Positions |
|---|---|---|---|---|---|
| 1 | NONE | | STANDARD | | 4 |

## Spectral Elements

| # | Short Filter | Long Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|
| 1 | F200W | F444W | BRIGHT2 | 10 | 1 | 4 | 4 | 858.942 | |

## Special Requirements

Sequence Observations 5, 6, Non-interruptible

## Observation

**Proposal 2562, Observation 6: HH30 - MIRI**

Tue Aug 29 22:01:08 GMT 2023

**Diagnostic Status: Error**

Observing Template: MIRI Imaging

## Diagnostics

(HH30 - MIRI (Obs 6)) Error (Form): Permission has not been granted for this program to use Special Requirement 'No Parallel Attachments'.

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (4) | HH-30-MIRI | RA: 04 31 37.4719 (67.9061329d)<br>Dec: +18 12 24.48 (18.20680d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |

*Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.*
*Category=Star*
*Description=[Circumstellar disks, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]*

## Template

| Subarray |
|---|
| BRIGHTSKY |

## Dithers

| # | Dither Type | Starting Point | Number of Points | Points | Starting Set | Number of Sets | Optimized For | Direction | Pattern Size |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 4-Point-Sets | | | | 4 | 1 | EXTENDED SOURCE | POSITIVE | DEFAULT |

## Spectral Elements

| # | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | F770W | FASTR1 | 80 | 4 | 1 | Dither 1 | 4 | 16 | 1117.942 | |
| 2 | F1280W | FASTR1 | 40 | 8 | 1 | Dither 1 | 4 | 32 | 1131.786 | |
| 3 | F2100W | FASTR1 | 20 | 16 | 1 | Dither 1 | 4 | 64 | 1159.475 | |

## Special Requirements

No Parallel Attachments

Sequence Observations 5, 6, Non-interruptible

## Observation

**Proposal 2562, Observation 15: HH30 - NIRCam - repeat**

Tue Aug 29 22:01:08 GMT 2023

**Diagnostic Status: Warning**

Observing Template: NIRCam Imaging

## Diagnostics

(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (3) | HH-30-NIRCAM | RA: 04 31 39.6595 (67.9152479d)<br>Dec: +18 13 5.42 (18.21817d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |

*Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.*
*Category=Star*
*Description=[Circumstellar disks, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]*

## Template

| Module | Subarray | Target Placement |
|---|---|---|
| B | FULL | Module Gap |

## Dithers

| # | Primary Dither Type | Primary Dithers | Subpixel Dither Type | Dither Size | Subpixel Positions |
|---|---|---|---|---|---|
| 1 | NONE | | STANDARD | | 4 |

## Spectral Elements

| # | Short Filter | Long Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|
| 1 | F200W | F444W | RAPID | 5 | 2 | 8 | 4 | 472.418 | |

## Special Requirements

Group Observations 15, 16, Non-interruptible

## Observation

**Proposal 2562, Observation 16: HH30 - MIRI - repeat**

Tue Aug 29 22:01:08 GMT 2023

**Diagnostic Status: Warning**

Observing Template: MIRI Imaging

## Diagnostics

(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (4) | HH-30-MIRI | RA: 04 31 37.4719 (67.9061329d)<br>Dec: +18 12 24.48 (18.20680d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |

*Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.*
*Category=Star*
*Description=[Circumstellar disks, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]*

## Template

| Subarray |
|---|
| BRIGHTSKY |

## Dithers

| # | Dither Type | Starting Point | Number of Points | Points | Starting Set | Number of Sets | Optimized For | Direction | Pattern Size |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 4-Point-Sets | | | | 4 | 1 | EXTENDED SOURCE | POSITIVE | DEFAULT |

## Spectral Elements

| # | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | F770W | FASTR1 | 80 | 4 | 1 | Dither 1 | 4 | 16 | 1117.942 | |
| 2 | F1280W | FASTR1 | 40 | 8 | 1 | Dither 1 | 4 | 32 | 1131.786 | |
| 3 | F2100W | FASTR1 | 20 | 16 | 1 | Dither 1 | 4 | 64 | 1159.475 | |

## Special Requirements

Group Observations 15, 16, Non-interruptible

## Observation

**Proposal 2562, Observation 7: Oph163131 - NIRCam**

Tue Aug 29 22:01:08 GMT 2023

**Diagnostic Status: Warning**

Observing Template: NIRCam Imaging

## Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (5) | 2MASS-J16313124-2426281 | RA: 16 31 31.2518 (247.8802158d)<br>Dec: -24 26 28.47 (-24.44124d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |

*Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.*

*K= 12.63*
*Category=Star*
*Description=[Circumstellar disks, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]*

## Template

| Module | Subarray | Target Placement |
|---|---|---|
| B | FULL | Module Gap |

## Dithers

| # | Primary Dither Type | Primary Dithers | Subpixel Dither Type | Dither Size | Subpixel Positions |
|---|---|---|---|---|---|
| 1 | NONE | | STANDARD | | 4 |

## Spectral Elements

| # | Short Filter | Long Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|
| 1 | F200W | F444W | BRIGHT2 | 3 | 3 | 12 | 4 | 858.942 | |

## Special Requirements

Offset -12.711611424643902 arcsec, -25.387873126628875 arcsec

Sequence Observations 7, 8, Non-interruptible

## Observation

**Proposal 2562, Observation 8: Oph163131 - MIRI**

Tue Aug 29 22:01:08 GMT 2023

**Diagnostic Status: Error**

Observing Template: MIRI Imaging

## Diagnostics

(Oph163131 - MIRI (Obs 8)) Error (Form): Permission has not been granted for this program to use Special Requirement 'No Parallel Attachments'.

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (5) | 2MASS-J16313124-2426281 | RA: 16 31 31.2518 (247.8802158d)<br>Dec: -24 26 28.47 (-24.44124d)<br>Equinox: J2000 | Epoch of Position: 2015.5 | |

*Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.*

*K= 12.63*
*Category=Star*
*Description=[Circumstellar disks, Pre-main sequence stars, Protoplanetary disks, Young stellar objects]*

## Template

| Subarray |
|---|
| BRIGHTSKY |

## Dithers

| # | Dither Type | Starting Point | Number of Points | Points | Starting Set | Number of Sets | Optimized For | Direction | Pattern Size |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 4-Point-Sets | | | | 4 | 1 | EXTENDED SOURCE | POSITIVE | DEFAULT |

## Spectral Elements

| # | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | F770W | FASTR1 | 60 | 4 | 1 | Dither 1 | 4 | 16 | 841.052 | |
| 2 | F1280W | FASTR1 | 30 | 8 | 1 | Dither 1 | 4 | 32 | 854.897 | |
| 3 | F2100W | FASTR1 | 5 | 32 | 1 | Dither 1 | 4 | 128 | 661.074 | |

## Special Requirements

No Parallel Attachments

Sequence Observations 7, 8, Non-interruptible